

Seamoon Pandey

B.E. Computer Engineering | AI & Backend Engineer

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Pokhara, Nepal | Date of birth: 27 January 2003 | Nationality: Nepali

PROFILE

Computer Engineering graduate of the Institute of Engineering, Tribhuvan University, applying for an M.Sc. in Computer Science. My final-year thesis paired a fine-tuned DistilBERT classifier with a polyglot microservice backend to detect cross-site scripting vulnerabilities. My interests lie in applied machine learning, honest model evaluation, and the distributed systems that serve models in production.

EDUCATION

Pashchimanchal Campus, Institute of Engineering, Tribhuvan University

Pokhara, Nepal

Bachelor of Engineering in Computer Engineering

2022 – 2026

Aggregate: **67.98% — First Division** | 1096.1 / 1612.5 marks | T.U. Regd. 3-2-45-173-2021

Tribhuvan University scale: Distinction $\geq 80\%$, First Division $\geq 65\%$, Second Division $\geq 50\%$. Examinations are set and moderated centrally by the university and marks are not scaled.

Thesis: RedSentinel — AI-Assisted Cross-Site Scripting (XSS) Vulnerability Scanner (Lead Architect, team of 3).

LANGUAGE PROFICIENCY

- **English** — IELTS Academic **7.5** overall, **CEFR C1** (15 August 2026). Listening 8.5 | Reading 7.5 | Writing 7.0 | Speaking 7.0. No band below 7.0.
- **Nepali** — Native speaker. **Medium of instruction:** English, throughout the Bachelor's degree.

THESIS PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner

Final-year Thesis, Team of 3 | [Code](#) | [Report](#) | [Demo](#)

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching. Reached near-perfect accuracy on an **8,869-sample** held-out split — a result I attribute to a deterministic, well-structured label space rather than to model strength, and report as such.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Architected a polyglot microservice system: a **TypeScript/NestJS** core orchestrating three **Python/FastAPI** AI services over a **Redis/BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritise payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend & Security:** React, Next.js, Flutter; XSS detection, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

SELECTED PROJECTS

- **AI Artifact Recognition Platform** (PEC Hackathon — **YOLOv8**, Flask, React) — Flask backend and React dashboard for real-time object recognition, with asynchronous inference and structured monitoring endpoints.
- **Community Connect** (OSM Hackathon — Express.js, Flutter) — full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.
- **Garva Diary** (BMC InnovateX — Ruby on Rails) — backend APIs on a Rails MVC architecture with structured validation workflows and mobile integration.

EXPERIENCE

Full-Stack Engineer — Lamida Inc.

Full-time | 23 Feb 2026 – Present

- Build scalable Web3 applications, including a decentralised autonomous organisation (DAO) for the tUSDT token.
- **Team Lead** — **Bitfolio**, from September 2026.

Freelance Full-Stack Developer

Contract, Remote

- Delivered production platforms on an **Express.js + React** stack — authentication, REST APIs, third-party integrations — while carrying a full course load: **Jyotisam**, an astrology consultation and booking platform (jyotisam.com), and **Mahadaan**, a donation and charity platform with secure data handling (mahadaan.com).

Mentor — Call for Enthusiasts Programme

Innovative Computer Engineering Students' Society (i-CES)

- Mentored junior students on scalable application design, clean backend architecture, and development practice.